2026 Korea-Finland Chip Design Symposium

When: Friday, September 11, 2026
13:00 ~ 18:00
Saturday, September 12, 2026
09:00 ~ 13:10

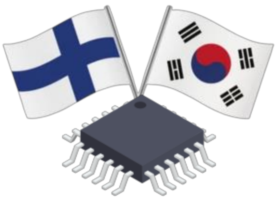
Organizer:
Prof. Kwantae Kim
Aalto University, Finland

Where: Aalto University
Odeion, Maarintie 8, 02150 Espoo, Finland

Host: Embassy of the Republic of Korea in Finland
Aalto University
Aalto Microelectronics Research Center (METKA)

Speakers

Speaker	Affiliation	Title	Country
Jussi Ryynänen	Aalto University	Professor	Finland
Kimmo Koli	Aalto University & Saab Finland	Research Fellow	Finland
Mikko Varonen	VTT	Research Team Leader	Finland
Abrar Akram	Tampere University	Assistant Professor	Finland
Praveen Buddireddy	Nokia	Technical Leader	Finland
SangEun Park	Spinverse	Project Manager	Finland
Jae-Yoon Sim	POSTECH	Professor	South Korea
Woo-Seok Choi	Seoul National University	Associate Professor	South Korea
Woo-Jun Choi	Yonsei University	Assistant Professor	South Korea
Minyoung Song	DGIST	Associate Professor	South Korea
Taekwang Jang	ETH Zurich	Associate Professor	Switzerland
Inhee Lee	University of Pittsburgh	Assistant Professor	USA
Soyeon Um	MIT	Postdoc	USA
Sohmyung Ha	New York University Abu Dhabi	Associate Professor	UAE



Friday, 11.Sep.2026

Time	Title	Speaker
12:30	Reception / Welcome Coffee	
	Session 1: Opening	
13:00	Introduction to the Symposium	**Prof. Kwantae Kim** Aalto University, Finland
13:15	Congratulatory Remarks	**Ambassador Jaewoong Lee** Embassy of the Republic of Korea in Finland
	Session 2: Circuits for Quantum Computing and AI	
13:30	Integrated Cryo-CMOS Circuit Systems for Quantum Computing	**Prof. Jae-Yoon Sim** POSTECH, South Korea
14:00	From Reactive to Predictive Power Management: The Evolution of Intelligent Digital Low-Dropout Regulators for Future AI Systems	**Prof. Abrar Akram** Tampere University, Finland
14:30	Energy-Efficient In-Memory Computing Processors for AI Acceleration	**Dr. Soyeon Um** MIT, USA
15:00	Break / Networking	
	Session 3: Toward High-Bandwidth, Energy-Efficient Chip Integration	
15:30	Connecting Korean and European Semiconductor Innovation: The Chips Ecosystem, Chips JU, and Opportunities for Collaboration	**SangEun Park** Spinverse, Finland
16:00	PLL-XO Codesign for Energy-Efficient and Ultra-Low Noise Frequency Synthesis	**Prof. Taekwang Jang** ETH Zurich, Switzerland
16:30	Break / Networking	
17:00	Energy-Efficient SoC Design through Pre-Silicon Power-Thermal Estimation Flows and Methodologies	**Praveen Buddireddy** Nokia, Finland
17:30	Enabling High-Bandwidth, Noise-Resilient Die-to-Die Interfaces	**Prof. Woo-Seok Choi** Seoul National University, South Korea
18:00	Banquet	

Saturday, 12.Sep.2026

Time	Title	Speaker
08:30	Reception / Welcome Coffee	
	Session 4: Wireless Transceivers from UWB to mmWave	
09:00	RF and Mixed Mode IC Design for Communications and Sensing	**Prof. Jussi Ryynänen** Aalto University, Finland
09:30	Low-Power and Compact IR-UWB Transmitters with Advanced Spectral Shaping	**Prof. Minyoung Song** DGIST, South Korea
10:00	Phased-Array MMICs and Modular 3D mmW & THz Packaging	**Dr. Mikko Varonen** VTT, Finland
10:30	Direct RF Sampling Receiver: Solution or Problem?	**Dr. Kimmo Koli** Aalto University & Saab Finland, Finland
11:00	Break / Networking	
	Session 5: Sensing Systems from Analog Front-Ends to mm-Scale Chips	
11:30	The World's Smallest Computer and Its Applications	**Prof. Inhee Lee** University of Pittsburgh, USA
12:00	Low-Power Architectures and Techniques for Bioimpedance Measurement	**Prof. Sohmyung Ha** New York University Abu Dhabi, UAE
12:30	Compact, Energy-Efficient BJT-Based Temperature Sensor with Active Sampling Noise Cancellation	**Prof. Woo-Jun Choi** Yonsei University, South Korea
13:00	Closing Remarks / Group Photo	**Prof. Kwantae Kim** Aalto University, Finland
13:10	Lunch	

Biography

Prof. Kwantae Kim, Aalto University, Finland Link

Organizer

Kwantae Kim received B.S., M.S., and Ph.D. degrees in the School of Electrical Engineering, KAIST, South Korea, in 2015, 2017, and 2021, respectively. He is an Assistant Professor at the Department of Electronics and Nanoengineering, School of Electrical Engineering, Aalto University, Finland. He was a Visiting Student in 2020, and a Postdoctoral Researcher from 2021 to 2023, at the Institute of Neuroinformatics, University of Zurich and ETH Zurich, and an Established Researcher from 2023 to 2024, at the Department of Information Technology and Electrical Engineering, ETH Zurich, Switzerland. His research interests include analog/mixed-signal ICs and full-custom memory ICs for intelligent audio systems, biomedical sensors, and flexible chips.

Prof. Jae-Yoon Sim, POSTECH, South Korea Link

"Integrated Cryo-CMOS Circuit Systems for Quantum Computing"

Jae-Yoon Sim received the B.S., M.S., and Ph.D. degrees from the POSTECH, in 1993, 1995, and 1999, respectively. From 1999 to 2005, he was a Senior Engineer with Samsung Electronics. Dr. Sim joined POSTECH in 2005 and is currently a professor. Since 2019, he has been the Director of the Scalable Quantum Computer Technology Platform Center. His research interests include sensor interface circuits, high-speed serial/parallel links, PLLs, data converters, AI accelerators and cryo-CMOS circuits for quantum computing. He received Takuo-Sugano Award in 2002, Jan Van Vessem Award in 2023 from ISSCC, and Scientist of the Month Award from Korea Ministry of Science and ICT in 2020. He had served on the Technical Program Committees for ISSCC, Symposium on VLSI Circuits, and ASSCC.

Prof. Muhammad Abrar Akram, Tampere University, Finland Link

"From Reactive to Predictive Power Management: The Evolution of Intelligent Digital Low-Dropout Regulators for Future AI Systems"

Muhammad Abrar Akram is a Tenure-Track Assistant Professor in the Faculty of Information Technology and Communication Sciences at Tampere University, Finland. He received his B.S. from the University of Punjab, Pakistan, and his combined M.S./Ph.D. from Kangwon National University, South Korea. He has held research and faculty positions at New York University Abu Dhabi and Qatar University. His research focuses on analog and mixed-signal CMOS integrated circuits for power management, with recent work on learning-enabled digital voltage regulators. He is a recipient of the President's Award and Company Special Award from the Korean Semiconductor Industry Association (KSIA), and the Best Paper Award at ICEIC 2021.

Dr. Soyeon Um, MIT, USA Link

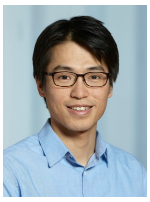
"Energy-Efficient In-Memory Computing Processors for AI Acceleration"

Soyeon Um received the B.S., M.S., and Ph.D. degrees from the School of Electrical Engineering, KAIST, Daejeon, South Korea, in 2020, 2021, and 2025, respectively. She is currently a Postdoctoral Associate at the Department of Electrical Engineering and Computer Science, MIT. Her current research interests include energy-efficient AI hardware design, with a particular focus on accelerators for deep neural networks, large language models and neuromorphic computing, as well as energy-efficient in-memory computing architectures.

SangEun Park, Spinverse, Finland

"Connecting Korean and European Semiconductor Innovation: The Chips Ecosystem, Chips JU, and Opportunities for Collaboration"

SangEun Park is a Project Manager at Spinverse, Finland, where she supports the preparation and management of collaborative European R&D and innovation projects. Her work focuses on Horizon Europe and Chips Joint Undertaking initiatives, with particular emphasis on consortium building, proposal development, and international collaboration in semiconductor and digital technology domains. She has extensive experience connecting Korean and European research and innovation ecosystems and supporting organisations in developing strategic partnerships for future technologies.



Prof. Taekwang Jang, ETH Zurich, Switzerland Link

"PLL-XO Codesign for Energy-Efficient and Ultra-Low Noise Frequency Synthesis"

Taekwang Jang received his B.S. and M.S. in electrical engineering from KAIST, Korea, in 2006 and 2008, respectively. In 2017, he received his Ph.D. from the University of Michigan. From 2008 to 2013, he worked at Samsung Electronics Company Ltd., Yongin, Korea. Currently, he is an associate professor at ETH Zürich and is leading the Energy-Efficient Circuits and Intelligent Systems group. His research focuses on circuits and systems for highly energy-constrained applications such as wireless sensor nodes and biomedical interfaces. Essential building blocks such as a sensor interface, energy harvester, power converter, communication transceiver, frequency synthesizer, and data converters are his primary interests. He received several awards from ISSCC and VLSI, including the 2024 IEEE SSCS New Frontier Award. He served as a TPC member for the ISSCC, an associate editor for the JSSC, and a distinguished lecturer for the SSCS.

Praveen Buddireddy, Nokia, Finland Link

"Energy-Efficient SoC Design through Pre-Silicon Power-Thermal Estimation Flows and Methodologies"

Praveen Kumar Buddireddy is a Technical Leader in the SoC Architecture and Specification group at Nokia, Finland. He received his B.Tech. degree in Computer Science from JNTU, India, in 2001. He has over 25 years of industry experience in CPU, SoC architecture, virtual platforms, system simulation, architecture exploration, and power-thermal analysis. Before Nokia, he worked at Texas Instruments, Qualcomm, and Intel, contributing to CPU and SoC modelling, platform-level analysis, and design methodology development. At Nokia, his work focuses on energy-efficient SoC design, especially pre-silicon power and thermal estimation flows and methodologies. He is also active in university collaboration initiatives in Finland and has authored or contributed to more than ten publications in industry technology forums, IC design symposiums, and EE Times. He received best paper awards at SNUG 2020 and Intel HSPE 2021, as well as the Most Innovative Paper Award at the Texas Instruments Technology Conference in 2011. He is also a passionate yoga instructor.

Prof. Woo-Seok Choi, Seoul National University, South Korea Link

"Enabling High-Bandwidth, Noise-Resilient Die-to-Die Interfaces"

Woo-Seok Choi received the B.S. and M.S. degree in electrical engineering and computer science from Seoul National University, Seoul, Korea, in 2008 and 2010, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Illinois at Urbana-Champaign, IL, USA, in 2017. From 2018 to 2019, he was a postdoctoral fellow at Harvard University, Cambridge, MA, USA. Since 2020, Woo-Seok has been with the Department of Electrical and Computer Engineering of Seoul National University, where he is currently an associate professor. His research interests include designing energy-efficient high-speed wireline transceivers and low-noise clocking circuits.

Prof. Jussi Ryynänen, Aalto University, Finland Link

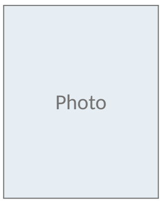
"RF and Mixed Mode IC Design for Communications and Sensing"

Jussi Ryynänen received the D.Sc. degree in electrical engineering from the Helsinki University of Technology, Espoo, Finland, in 2004. He is currently a full professor and the Dean of the School of Electrical Engineering, Aalto University. He has authored or co-authored more than 200 refereed journal and conference papers in analog and RF circuit design. He is currently SG Member for the European Solid-State Circuits Conference (ESSCIRC) and the IEEE Nordic Circuits and Systems Conference (NORCAS). He has served as TPC member of IEEE ISSCC and ESSCIRC.

Prof. Minyoung Song, DGIST, South Korea Link

"Low-Power and Compact IR-UWB Transmitters with Advanced Spectral Shaping"

Minyoung Song received the B.S. and Ph.D. degrees in Electrical Engineering from Korea University, Republic of Korea, in 2006 and 2013, respectively. He is currently an Associate Professor with the Department of Electrical Engineering and Computer Science at DGIST, Korea. Before joining DGIST in 2023, he was a Senior Engineer at Samsung Electronics, where he developed analog and mixed-signal ICs. He then joined imec in the Netherlands, where he led the design of energy-efficient RFICs for biomedical and IoT applications. His current research focuses on analog, mixed-signal, and RF IC design for wireless communications and radar, as well as cryogenic CMOS circuits and interfaces for quantum sensing. He currently serves on the Technical Program Committee of ISSCC.



Dr. Mikko Varonen, VTT, Finland

"Phased-Array MMICs and Modular 3D mmW & THz Packaging"

[Bio TBD]

Dr. Kimmo Koli, Aalto University & Saab Finland, Finland

"Direct RF Sampling Receiver: Solution or Problem?"

Kimmo Koli received the M.Sc., Lic. Tech., and D.Sc. degrees in electrical engineering from the Helsinki University of Technology (now Aalto University), Helsinki, Finland, in 1991, 1995, and 2000, respectively. From 1989 to 2000, he was a Research Engineer with the Electronic Circuit Design Laboratory, Helsinki University of Technology. From 2000 to 2008, he was with the Nokia Research Center, Helsinki, working on audio and sensor interfaces and wireless transceivers. From 2008 to 2014, he was working with STMicroelectronics (later ST-NXP Wireless, ST-Ericsson, and Ericsson) and, from 2014 to 2024, with HiSilicon at Huawei Technologies Oy (Finland) Company Ltd., Helsinki, focusing on RF and AMS design and research for wireless and power management applications. He is currently both with the Department of Electronics and Nanoengineering, Aalto University, and Saab Finland Oy focusing on RF and AMS design for wireless sensing and communications applications. He has authored or coauthored over 40 journal articles and conference papers and holds 19 granted or pending patents. He served as a member for the Technical Program Committee for IEEE ISSCC from 2011 to 2014 and IEEE ESSCIRC from 2011 to 2019.

Prof. Inhee Lee, University of Pittsburgh, USA **Link**

"The World's Smallest Computer and Its Applications"

Inhee Lee earned his B.S. and M.S. in electrical and electronic engineering from Yonsei University in 2006 and 2008, respectively. He obtained his Ph.D. in electrical and electronic engineering from the University of Michigan in 2014. From 2015 to 2019, he served as an assistant research scientist at the University of Michigan. In 2019, he became an assistant professor at the University of Pittsburgh. His research focuses on developing millimeter-scale or smaller sensing/computing systems for ecological, biomedical, and AI-of-things applications. Dr. Lee has served as a TPC Member for IEEE ISSCC, VLSI Symposium, CICC, A-SSCC, and CAS Analog Signal Processing.

Prof. Sohmyung Ha, New York University Abu Dhabi, UAE **Link**

"Low-Power Architectures and Techniques for Bioimpedance Measurement"

Sohmyung Ha received the B.S (summa cum laude) and the M.S. degrees in Electrical Engineering from KAIST, Daejeon, Korea, in 2004 and 2006, respectively. From 2006 to 2010, he worked as an integrated circuit designer at Samsung Electronics. He then returned to academia as a Fulbright Scholar and earned the M.S. and Ph.D. degrees in Bioengineering from the University of California, San Diego, La Jolla, CA, USA, where he received the Engelson Best Ph.D. Thesis Award in Biomedical Engineering. Since 2016, he has been with New York University Abu Dhabi, Abu Dhabi, UAE, where he is currently an Associate Professor of Electrical Engineering and Bioengineering.

He currently serves as an Associate Editor of the IEEE Open Journal of the Solid-State Circuits Society and previously served as an Associate Editor of the IEEE Transactions on Biomedical Circuits and Systems. He is a member of the Analog Signal Processing Technical Committee and the Biomedical and Life Science Circuits and Systems Technical Committee of the IEEE CASS. He previously served on the ITPC of the IEEE ISSCC and is currently the TPC Co-Chair of the IEEE BioCAS 2026.

Prof. Woo-Jun Choi, Yonsei University, South Korea **Link**

"Compact, Energy-Efficient BJT-Based Temperature Sensor with Active Sampling Noise Cancellation"

Woojun Choi is an Assistant Professor in the Department of Integrated Display Engineering at Yonsei University, Seoul, Korea. He received his B.S. and Ph.D. degrees in Electrical and Electronic Engineering from Yonsei University. Prior to joining Yonsei University, he was an Assistant Professor at Kyung Hee University and a Postdoctoral Research Fellow at ETH Zürich. His research interests include high-performance and energy-efficient analog and mixed-signal integrated circuits for display driver ICs, biomedical systems, neural interfaces, and smart sensor applications. He received the IEEE ISSCC 2021 Takuo Sugano Award for Outstanding Far-East Paper and the IEEE SSCS Predoctoral Achievement Award.

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Tiny Systems and Circuits (TSirc) Group, Aalto University